

Previous Class (Sept 22)

- ✓ 1) Problem Statement
- ✓ 2) Geometric Intuition of SVM
- ✓ 3) Hard Margin SVM
- ✓ 4) Soft Margin SVM
- ✓ 5) Algebraic Intuition of SVM
- ✓ 6) Hinge loss
- ✓ 7) SVM Imbalance
- ✓ 8) Code implementation

Today's class

- 1) Recap
- 2) Dual Formulation of SVM
- 3) Runtime of SVM
- 4) Kernel SVM (non-linear SVM)
- 5) Code walk through
- 6) Impact of outliers
- 7) Doubt Solving session (11:15 pm)

Problem Solving session → 2nd, 4th, 6th Oct

$x_1 \quad x_2 \quad x_3$

$$\begin{aligned} \rightarrow & \alpha_1 \alpha_1 y_1 y_1 x_1^T x_1 + \alpha_1 \alpha_2 y_1 y_2 x_1^T x_2 \\ & + \alpha_1 \alpha_3 y_1 y_3 x_1^T x_3 \\ & + \alpha_2 \alpha_1 y_2 y_1 x_2^T x_1 + \dots \end{aligned}$$

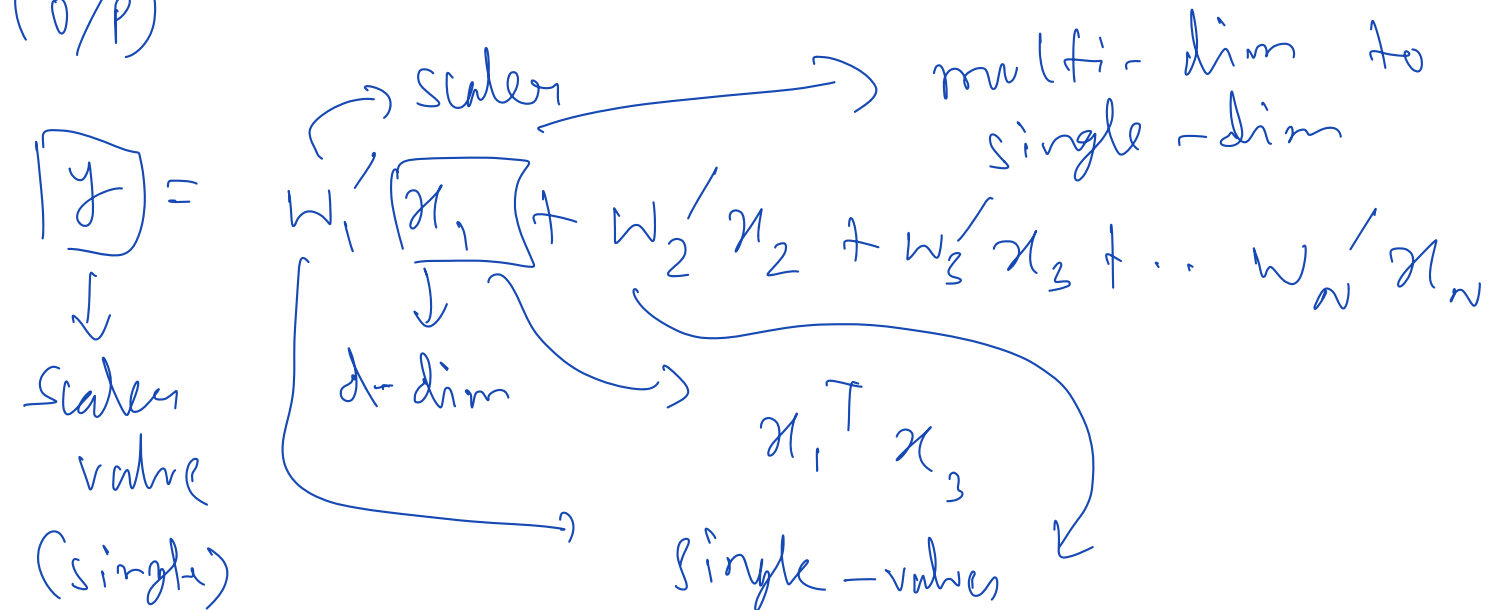
N data points

	F_1	F_2	F_3	$\dots$	F_d
x_1					
x_2					
x_3					
$\vdots$					
x_N					

d -dim

(o/p)

$$y = w_1 F_1 + w_2 F_2 + w_3 F_3 + \dots + w_d F_d$$



$$\boxed{x_1} = \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \end{bmatrix}, \quad \boxed{x_2} = \begin{bmatrix} x_{21} \\ x_{22} \\ x_{23} \end{bmatrix}$$

$$\begin{aligned} \overbrace{x_1^T x_2} &= \begin{bmatrix} x_{11} & x_{12} & x_{13} \end{bmatrix} \begin{bmatrix} x_{21} \\ x_{22} \\ x_{23} \end{bmatrix} \\ &= x_{11} \cdot x_{21} + x_{12} \cdot x_{22} + x_{13} \cdot x_{23} \\ &= \text{single-value} \end{aligned}$$

x_1, x_2 separately

$$\boxed{x_1^T x_2} \quad \checkmark$$

scaler values

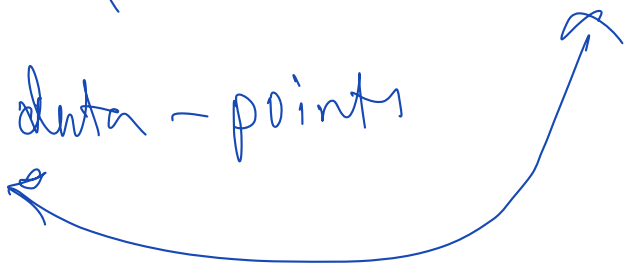
$$\boxed{x_1, x_2}$$

→

1000

dim

100 data-points



num. of data
(n)

$O(\underline{n} \times \underline{d})$ space
dim

$$x_1^T x_2 \rightarrow O(n \times 2) \text{ space}$$

If $d \gg n$, prefer $O(n^2)$ space

If $n \gg d$, prefer $O(nd)$ space

10:20 resume

$$x_1 \rightarrow \begin{bmatrix} x_{11} & x_{12} \end{bmatrix} \rightarrow x_1' \quad \text{degree 2}$$

$$\rightarrow x_{11}^2, x_{12}^2, x_{11} \cdot x_{12}$$

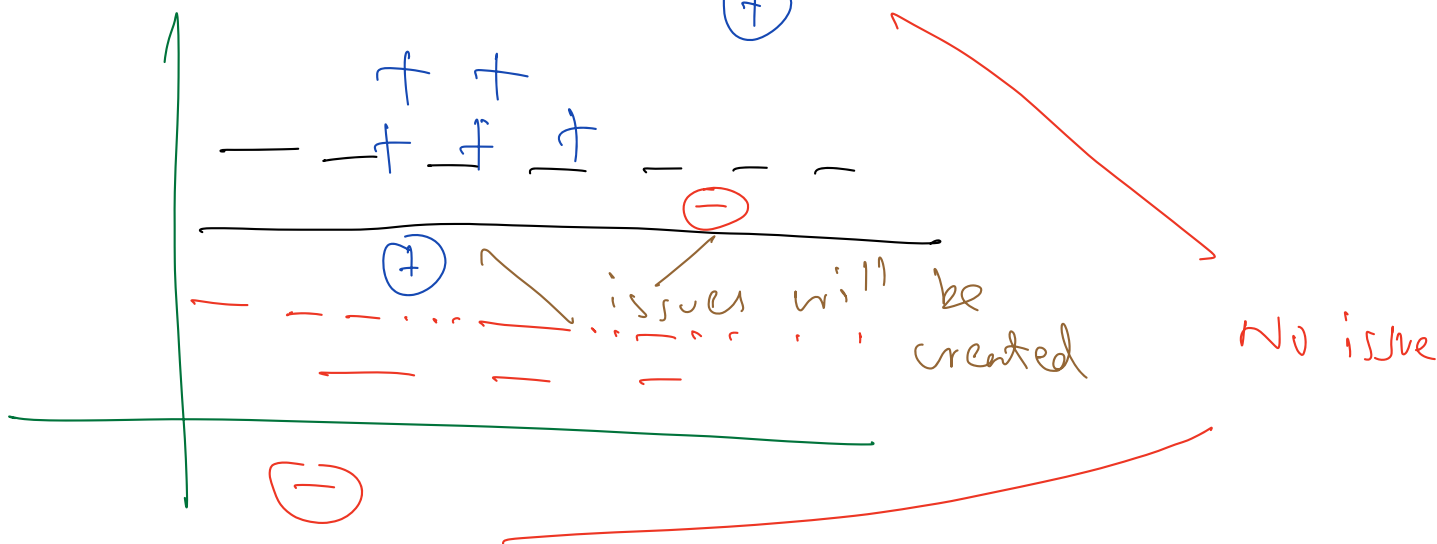
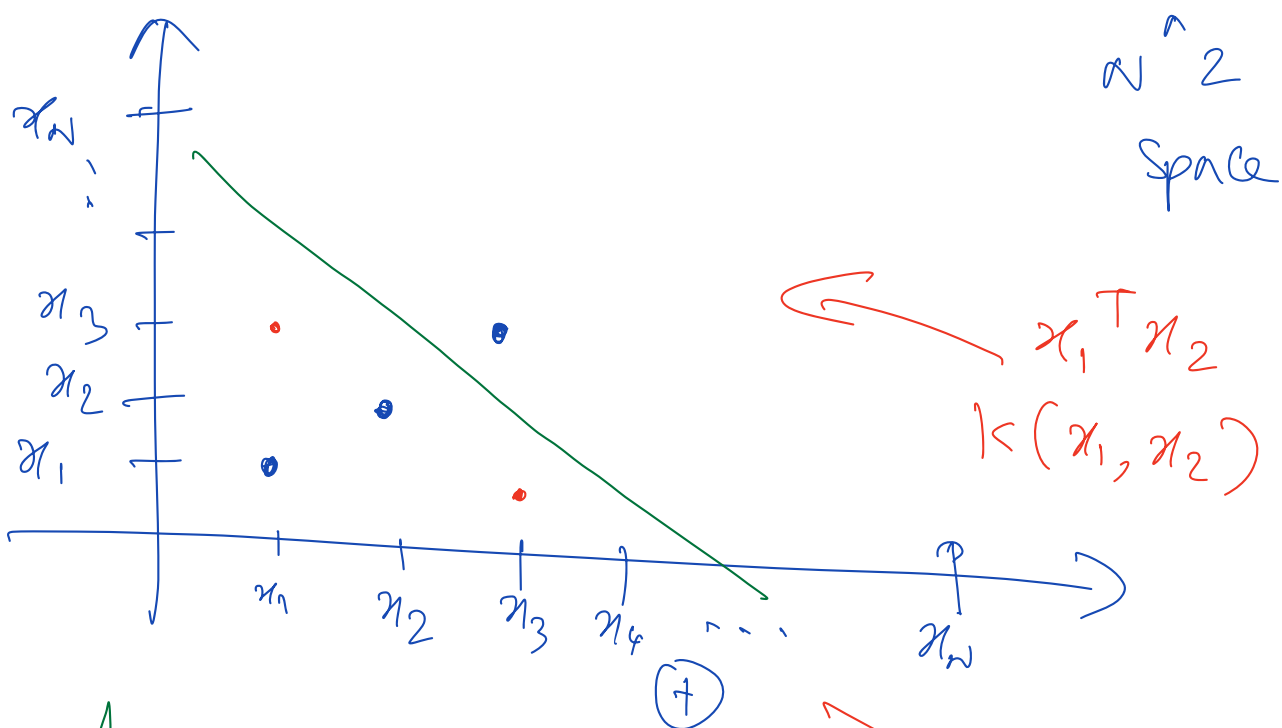
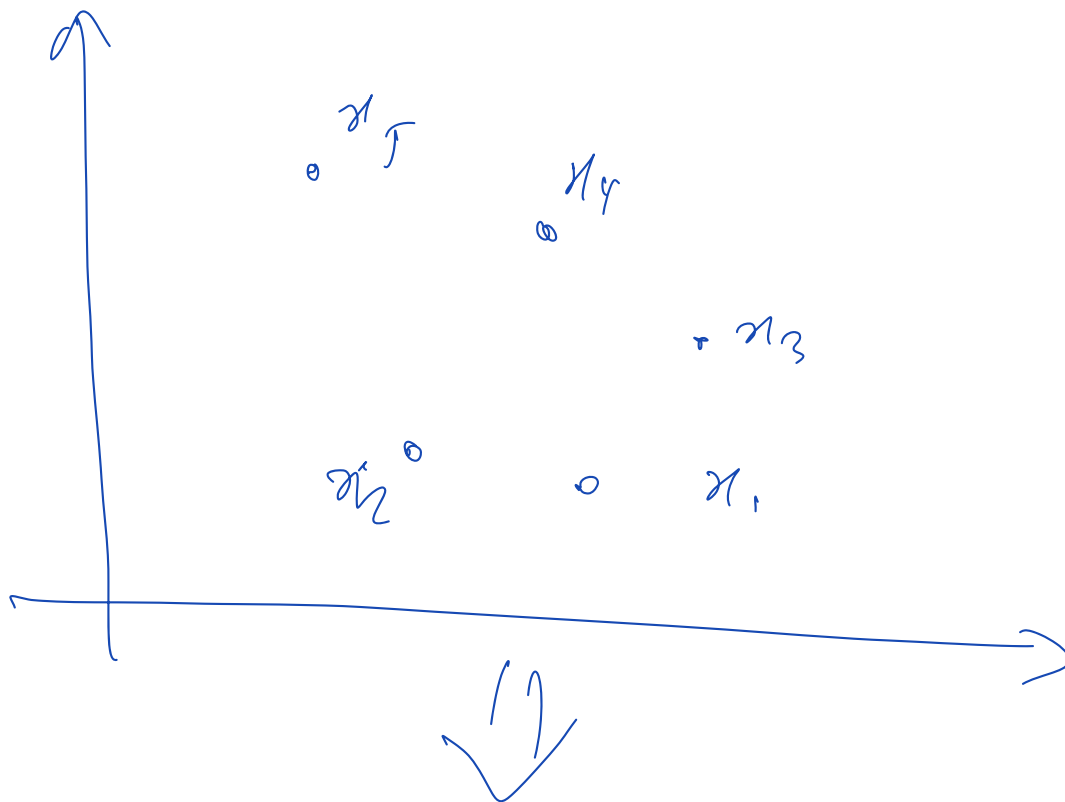
$$\rightarrow x_{11}^3, x_{12}^3, x_{11}^2 x_{12}, x_{11} \cdot x_{12}^2$$

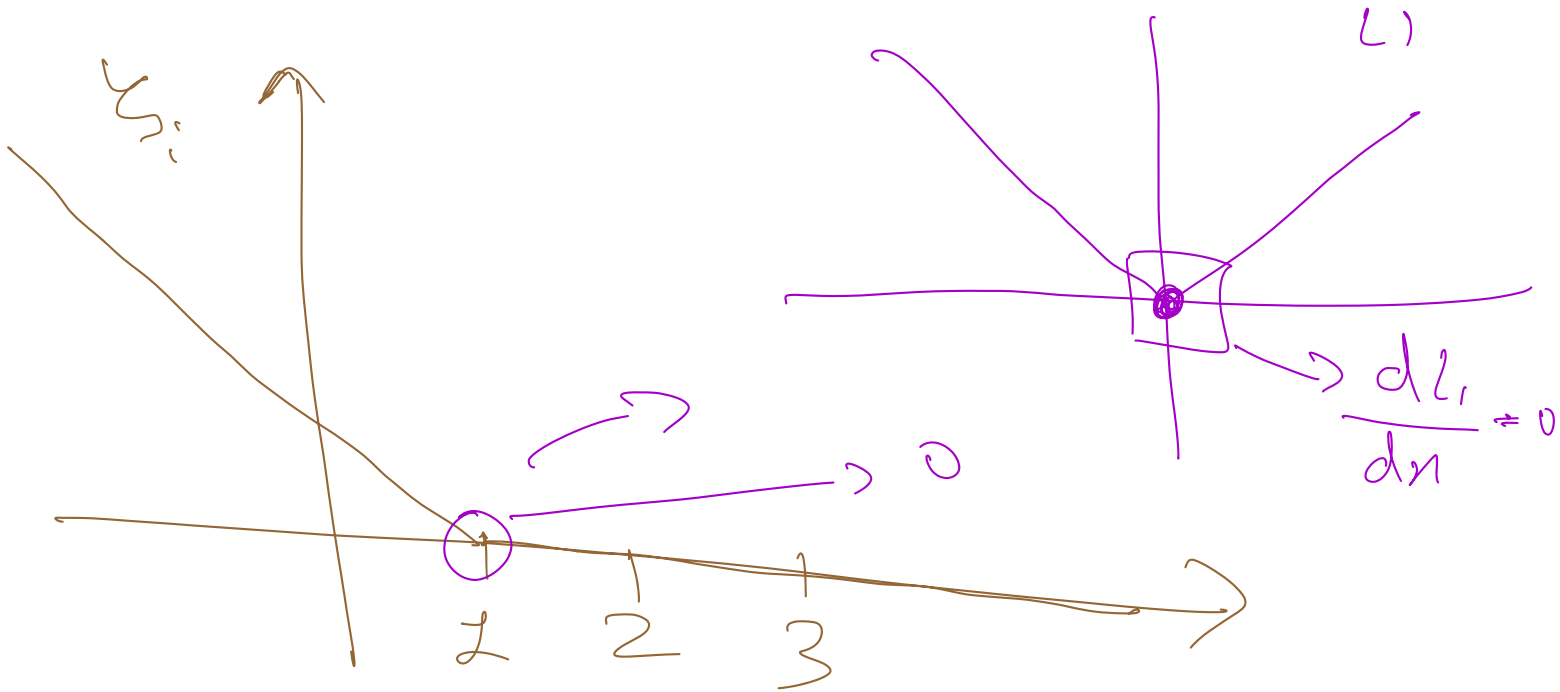
$$x_1^T \cdot x_2 \rightarrow \text{degree 1}$$

$$x_1'^T \cdot x_2'$$

similarity in higher-dim

$$k_2(x_1, x_2) \rightarrow \begin{matrix} x_1 \rightarrow x_1' \\ x_2 \rightarrow x_2' \end{matrix} \rightarrow x_1'^T x_2'$$





$$z = y_i (w^T x_i + b)$$